Uniwersytet SWPS

Task list no. 1 – Execute and observe runtime differences between algorithms of O(n), O(n²), and O(log n), Measure and record algorithm performance for different input sizes.

Task 1. Write three functions with different time complexities, and we want to compare experimental results to theoretical expectations.

1. Linear search (O(n))
2. Insertion sort ($O(n^2)$)
3. Binary search (O(log n))

Task 2. Run these functions and note the printed results (execution time for each n) in a table.

n	*Linear search* $(O(n))$	*Insertion sort*$(O(n^2)$	*Binary search*$(O(\log n))$
1000			
5000			
10000			
50000			
100000			
500000			
1000000			

The lab report should contain:

1. your code along with a table of runtimes for n = 1,000- 5,000-10,000-50,000-100,000,500,000,1,000,000

2. Interpret the relationship between input size and runtime growth.

Note. ALL PROGRAMS SHOULD COMPILE WITHOUT ERROR AND WORK (i.e. DO EXACTLY WHAT IS DESCRIBED IN THE TASK)!!!

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